

Peak electrical power of a small vertical-axis wind turbine with 3D-printed textured blades

Wind-tunnel tests of one plain and three fuzzy-skin blade sets,
with rotor speed and surface roughness

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Abstract

Fuzzy skin is a slicer setting that displaces the outer wall of a 3D print by a random distance up to a set thickness [1]; it changes a printed blade's surface texture without changing its CAD model. If texture changes a printed rotor's output, print settings must be controlled whenever rotors are compared. Roughness degrades the lift of airfoil sections [2] and lowers the Reynolds number at which a circular cylinder's drag falls [3]; neither result predicts its effect on the thin curved-plate blades of the small vertical-axis rotor studied here. We tested this rotor in the URI wind tunnel on 1 October 2026, fitted in turn with four blade sets: without fuzzy skin (Plain) and with fuzzy-skin thicknesses of 0.050 mm, 0.101 mm and 0.202 mm (FS 0.05, FS 0.10, FS 0.20). Each rotor was swept twice through 14 fan speeds (10.2–38.0 m/s by the tunnel calibration) with an automated constant-current load; light-load rotor speed and surface roughness were also measured. Averaged over wind speed, FS 0.10 and FS 0.20 produced +12.8% [+10.1, +15.5] and +16.2% [+13.5, +19.0] more peak electrical power than Plain (Tukey simultaneous 95% intervals); the FS 0.05 difference, +1.8% [−0.6, +4.2], was not resolved, its interval including zero. The gains were largest at 21–25 m/s. Averaged over wind speed, they came with a higher extrapolated open-circuit voltage and no resolved change in apparent source resistance; where they were largest, a lower source resistance also contributed. At light load, averaged over 12.4–38.0 m/s, FS 0.10 and FS 0.20 turned +6.6% and +9.1% faster than Plain, with no resolved difference in voltage per rpm; we infer that the gains arise in the rotor, not in the generator. Measured roughness (R_a , P_a) did not rank the sets by power. Each setting was one print, mounted once and tested in order of increasing thickness, and Plain also has a coarser layer period (0.20 mm against 0.10 mm), so attributing the differences to fuzzy skin requires replicate prints, remounting and an interleaved run order. Until then, rotors compared on this rig, including in the planned fault tests, should be printed from one recorded set of print settings.

1 Introduction

Printed blades carry surface texture even without fuzzy skin: the Plain set tested here has 0.20 mm layer lines (Section 6.6). If texture changes rotor output, print settings must be controlled in any experiment that compares rotors, including the fault tests planned for this rig (Section 10.1), and texture becomes a design variable. Fuzzy skin makes the texture adjustable without changing the

CAD geometry. In PrusaSlicer it resamples each outer-wall path with a random step size and shifts each new point inside or outside the wall by a random distance up to the fuzzy-skin thickness [1]; the Bambu Studio project used for these blades stores the same two parameters, thickness and point distance (Section 2.2).

On airfoil blade sections at Reynolds numbers of $1.6\text{--}4.0 \times 10^6$, increasing roughness height and density decreases the maximum lift, lift-curve slope and lift-to-drag ratio [2]. On a circular cylinder, roughness lowers the critical Reynolds number, at which the drag falls, leaves the subcritical drag unchanged and raises the transcritical drag [3]. The blades tested here are thin curved plates with square edges at chord Reynolds numbers of 3.2×10^4 to 1.2×10^5 (Section 2.1). Neither result predicts the sign or size of a texture effect on such blades, so it was measured.

One rotor was fitted in turn with the four blade sets and tested at 14 wind speeds, twice per set, to answer four questions:

1. Does peak electrical power differ between the sets? Tukey simultaneous intervals on run means [4] (Section 6.2; conclusion 1).
2. Do the differences depend on wind speed? Rotor $\times$ set point interaction (Section 6.3; conclusion 2).
3. Does a difference arise in the rotor or in the generator? A Thévenin fit of each load ladder, light-load rotor speed and voltage per rpm (Sections 6.4 and 6.5; conclusion 3).
4. Does measured surface roughness order the sets by power? Profilometer Pa and Ra (Section 6.6; conclusion 4).

The report covers the 1 October 2026 session only; earlier laboratory (June–July) and rig (August–September) tests are outside its scope. A fifth blade set run that day is excluded because its run log records its texture as unknown. The report also responds to two earlier requests: blade identity (Table 1) and an outline for a fault-testing plan (Section 10.1). Data and code accompany it (Appendix E).

2 Test articles

2.1 Rotor

The rotor has three straight, untwisted blades attached at $R = 0.1016\text{ m}$ from the axis, with span $H = 0.2451\text{ m}$. The blade section is a thin curved plate that turns through 183° (48 mm chord, 24.5 mm depth, 1.79 mm wall, square edges); it is not an airfoil. These are design values; the printed blades were not measured. The outer swept radius is larger than R and was not documented, so the nominal swept area $A = 2RH = 0.0498\text{ m}^2$ is used only to normalise power. Chord Reynolds numbers use the wind speed and assume air at 20°C ; air temperature was not recorded.

2.2 Blade sets

Table 1 lists the sets. A set mounted on the rig is called a rotor and named after the set. According to the project’s correspondence, the College of Engineering’s I²(s) Print Lab printed all sets from one CAD model of the blade. By design they differ in fuzzy-skin thickness; Plain also differs in layer period. The earlier labels identify the files: Plain’s runs are labelled smooth and its scan baseline, and Ra 20/40/80 are names, not measured Ra values.

The Print Lab’s slicer project (`turbine_default.3mf`, Bambu Studio, saved 17 August) holds three single-blade plates ($24.5 \times 48.0 \times 245.1\text{ mm}$); two match FS 0.10 and FS 0.20, and none matches FS 0.05. It specifies a Bambu Lab P1S with a 0.2 mm nozzle, 0.10 mm layers, PLA and four wall

Table 1. Blade sets. Layer period: dominant spanwise wavelength of the measured surface (Appendix C); —: no spectral peak between 60 and 320 μm .

Rotor	Fuzzy skin	Earlier label	Layer period	Print record
Plain	none	smooth, baseline	0.20 mm	No slicer record; the layer period indicates 0.2 mm layers.
FS 0.05	0.050 mm	Ra 20	0.10 mm	Not in the slicer project; thickness from the 1 October run log.
FS 0.10	0.101 mm	Ra 40	0.10 mm	Matches plate “40 RA” of the slicer project.
FS 0.20	0.202 mm	Ra 80	—	Matches plate “80 RA” of the slicer project.

loops. Fuzzy skin is painted on both faces of each blade at a 0.2 mm point distance and covers 96.5% of the surface; the square edges, end caps and two mounting-hole bores are unpainted. Only the fuzzy-skin thickness differs between plates. The project does not record which prints were made from it, and the run logs record only the thickness, so its other settings are not confirmed for the printed sets. Of these settings, only the layer height can be checked: the 0.10 mm layer period measured on FS 0.05 and FS 0.10 matches it.

2.3 Surface measurement

Each set was scanned on 1 October before the runs (13:07–13:22) with a Keyence VR-6000 optical profilometer (VR-6100 head, $\times 160$): one field of about 2.0×1.4 mm on one blade per set, two on FS 0.20, at $1.85 \mu\text{m}$ per pixel (Fig. 1). Field locations on the blades were not recorded. Each spanwise profile (across the layer lines) is levelled. Pa is the arithmetic mean deviation of the levelled profile; Ra is that of the roughness profile, the levelled profile minus its Gaussian mean line (ISO 16610-21 [5], $\lambda_c = 0.25$ mm), over a 1.13–1.17 mm evaluation length, shorter than the default of five cut-off lengths (1.25 mm) [6]; Appendix C gives the details. Each value is the mean over a field’s pixel columns; adjacent columns are correlated, so the field is the sampling unit.

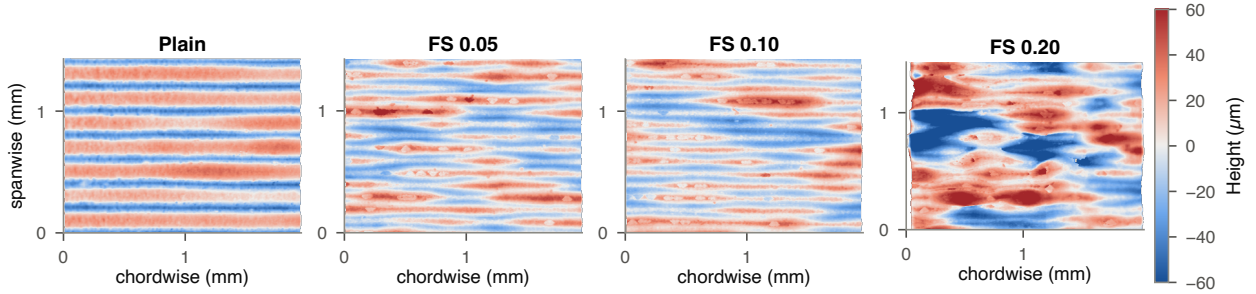


Figure 1. Profilometer height maps, one field per blade set, each column levelled. Layer lines run horizontally (chordwise); the Pa and Ra profiles run vertically (spanwise). Colour is clipped at $\pm 60 \mu\text{m}$, which FS 0.20 exceeds over 8% of its area.

3 Facility and instrumentation

3.1 Wind speed

The tunnel fan is driven by an ABB ACS550 variable-frequency drive, commanded in rpm by a Portenta Machine Control over Modbus (Fig. 2). Wind speed v is calculated from the commanded fan speed N with the tunnel’s calibration (Test 1, 13 February 2026), a least-squares line through eleven points (Appendix D):

$$v = 0.02132 N - 0.424 \quad (\text{m/s}; N \text{ in rpm}).$$

Of these points, eight were measured, from 0 to 700 rpm (up to 14.0 m/s); the calibration file marks the other three (1400–2400 rpm) as possibly read from the trend line of the original record, and the Test 1 instrument is not documented. The wind speeds assigned to the eleven set points above 700 rpm (16.6–38.0 m/s) therefore rest on the assumption that air speed stays linear in fan speed. At 500–700 rpm the line is 2.8–4.6% above the measured speeds; a line through the measured points alone gives 36.0 m/s at 1800 rpm, 5% below the adopted value. The calibration affects absolute wind speed and $C_{P,\text{el}}$, not the comparison between rotors, which are compared at identical fan speed.

The median fan-motor current at each set point agreed across the eight runs within 0.1 A, the drive’s reading resolution (13.6 A at 1800 rpm), so the fan operating point did not change measurably. Air temperature and pressure were not recorded. No blockage correction is applied; blockage is assumed to act alike on rotors of one geometry.

3.2 Electrical measurement

The rotor shaft drives a three-phase generator whose output passes through a three-phase bridge rectifier to a Chroma 63004-150-60 DC electronic load in constant-current mode. The generator’s make, model and winding resistance were not recorded. A Python host sets the load current and records the terminal voltage V and current I that the load reports at the end of each dwell; electrical power is $P = VI$. All rotors were measured with the same instrument at the same settings, so an instrument gain error cancels in the ratios between rotors.

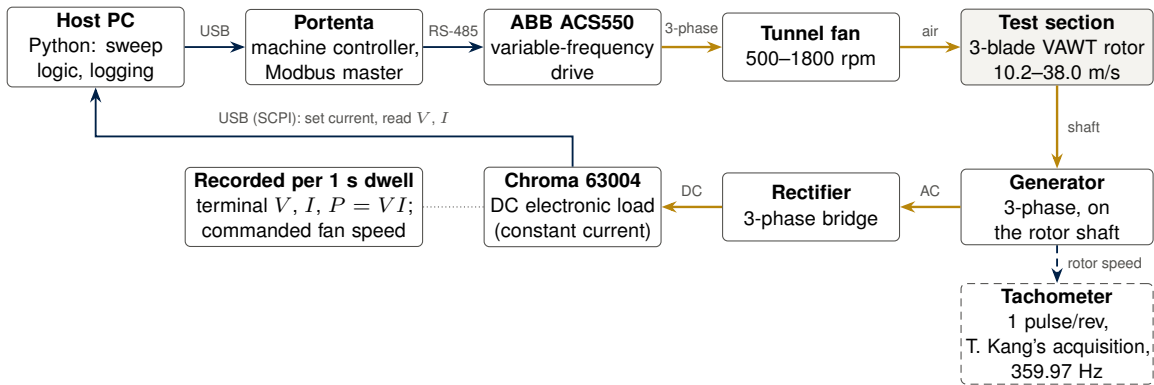


Figure 2. Measurement and control chain. The dashed path is T. Kang’s separate rotor-speed acquisition.

3.3 Rotor speed

The rig’s rotor-speed input counted no pulses on 1 October. T. Kang recorded rotor speed on a separate data acquisition system from a tachometer signal sampled at $f_s = 359.97$ Hz; the sensor type and target were not recorded. His script (RPM.m, in the data package) treats each pulse as one revolution, splits each run into 14 segments at rises in the 5 s mean speed, assigns them to the set points in order, and reports for each the most frequent per-revolution speed n_0 , rounded to 1 rpm. The summaries carry no time stamps. Because a pulse interval is a whole number of samples, n_0 is resolved to the larger of 1 rpm and $n_0^2/(60f_s)$, that is 0.7–4.0% of n_0 .

We take n_0 as the light-load speed, the rotor speed with the generator unloaded: the load is held at 0 A while the fan settles (Section 4), and the rotor then turns at its highest and steadiest speed of the set point. From 600 rpm the rig’s record supports this and the matching of set points. The terminal voltage at the first load step (1–10 mA), V_1 , follows the least-squares line $V_1 = 32.1 \text{ mV/rpm} \times n_0 - 0.22 \text{ V}$ over the 104 rotor, run and set-point values (Fig. 8c), and V_1/n_0 averages 31.5 mV/rpm with a coefficient of variation of 1.7%. At the first set point, 500 rpm, the fan settles with the load at 10 mA instead, and the speed that V_1 implies on this line exceeds n_0 by 3–12%; 500 rpm is therefore not used for speed. Rotor speed at each load dwell was not recorded.

4 Test procedure

4.1 Sweep protocol

Each run (sweep) visits 14 fan set points from 500 to 1800 rpm in 100 rpm steps. At each set point the host waits until the fan speed is within the larger of 10 rpm and 2% of the command, then until the generator’s terminal voltage changes by no more than 1% (or 0.02 V, whichever is larger) over each of three successive 0.5 s intervals; the voltage wait lasts at least 2 s, and each wait at most 20 s. While the fan settles the load is at 0 A, except at the first set point, where it stays at the 10 mA set before the fan starts. The load then raises its current demand from half a step, in steps of 20 mA at 1800 rpm scaled with v^2 at lower speeds (minimum 2 mA), dwelling 1 s per step, and stops after two consecutive dwells at or below 80% of the running maximum. The resulting ladders of 13–30 dwells trace the power–current curve through its maximum (Fig. 3). A run takes about 8 minutes; all runs used the same protocol (identifier 94bed28333f7).

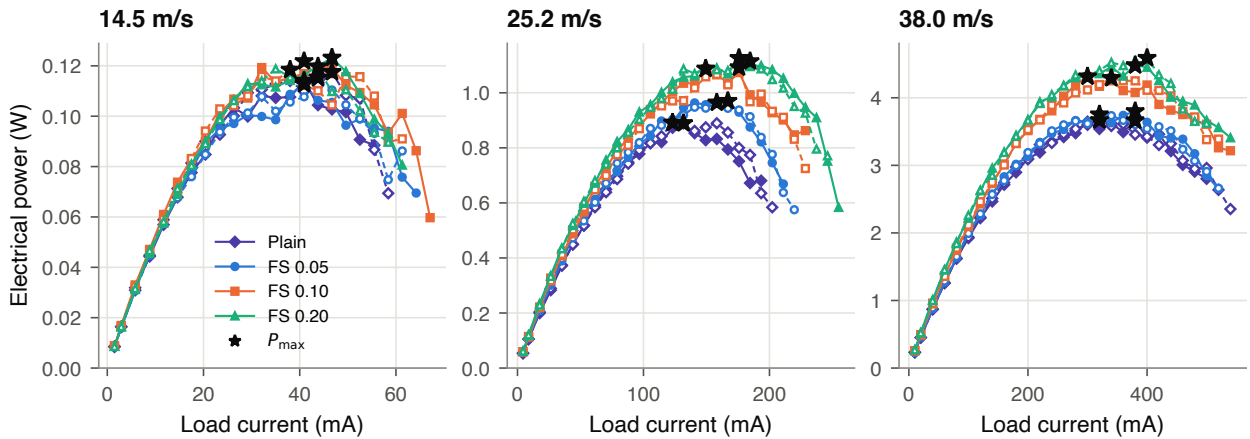


Figure 3. Measured load ladders at three wind speeds, all eight runs (solid: run 1; dashed, hollow: run 2). Stars mark $P_{\max}$, the largest dwell power.

4.2 Session

Each rotor was mounted once and swept twice without removal (Table 2), in the order Plain, FS 0.05, FS 0.10, FS 0.20, from 15:50 to 18:24. The two Plain runs started 60 minutes apart; the other pairs ran back to back.

Table 2. Runs of 1 October 2026: local time of the first and last dwell, and the run mean of $P_{\max}$ relative to the Plain rotor (Section 5).

Rotor	Run	Time	Change (%)	Data file stem
Plain	1	15:50–15:57	−0.4	v1_smooth_20261001
	2	16:50–16:58	+0.4	v1_smooth_repeat_20261001
FS 0.05	1	17:18–17:25	+2.3	v1_Ra20_20261001
	2	17:27–17:34	+1.3	v1_Ra20_repeat_20261001
FS 0.10	1	17:42–17:50	+12.4	v1_Ra40_20261001
	2	17:51–17:59	+13.1	v1_Ra40_repeat_20261001
FS 0.20	1	18:07–18:15	+15.7	v1_Ra80_20261001
	2	18:16–18:24	+16.8	v1_Ra80_repeat_20261001

4.3 Data quality

Of the 112 ladders, 110 stopped on the power roll-off. The other two stopped after the peak on the host’s voltage floor: the terminal voltage fell below 0.5 V on two consecutive dwells while the current still followed the demand. In these two ladders, Plain run 1 at 500 rpm (49%) and FS 0.05 run 2 at 600 rpm (57%), the last dwell above the floor had already fallen below 80% of the maximum (percentages in parentheses), so the peak lies inside every ladder. Peak powers recomputed from the dwell records equal the values logged by the rig. One Plain blade broke at the end of run 2; at the last set point, 1800 rpm, $P_{\max}$ agrees with run 1 within 0.1%. One FS 0.10 blade carried a small sensor magnet in an earlier test; whether it was fitted on 1 October was not recorded.

5 Data reduction and statistical analysis

Peak power. $P_{\max}$ is the largest dwell power in a ladder. A least-squares parabola through the maximum and up to two dwells on either side (the rig software’s rule) gives a second estimate; in ten ladders the fit is not concave or its vertex lies outside those dwells, and the maximum is used. The two estimators resolve the same pairs (Table 3). Because dwells last 1 s and speed was not recorded per dwell, rotor equilibrium within a dwell is not shown. At 500 rpm, which settles with the load at 10 mA, the terminal voltage rose between the first two dwells in seven of eight runs, so the rotor was still accelerating when the ladder began. $P_{\max}$ is the peak of this ramp protocol.

Rotor comparison. Differences are analysed as ratios, on $\ln P_{\max}$. For each run, y is the mean over the 14 set points of $\ln(P_{\max}/P_{\text{ref}})$, with P_{ref} the geometric mean of the two Plain runs at that set point, so each wind speed carries equal weight. The pooled run-to-run standard deviation of y is $\sigma = 0.58\%$ (4 degrees of freedom). The rotor effect is tested by one-way analysis of variance (ANOVA) on the eight run means, and all six pairs are compared with Tukey’s honestly significant difference procedure (simultaneous 95% intervals; studentized range $q_{0.05;4,4} = 5.76$) [4]. A difference is resolved when its interval excludes zero. With two runs per rotor the p -values rest on a normal model and are reported down to 0.001; an exact permutation test of the rotor effect gives $p = 1/105$,

the smallest attainable. Because each rotor’s two runs were consecutive, neither test separates a rotor effect from drift (Section 8).

Mounting and drift. The two runs of a rotor share its print and mounting, so the intervals measure repeatability on one mounting. Mounting-to-mounting variation τ , the standard deviation of a rotor’s mean power between mountings, was not measured. For each resolved difference the largest τ at which it would stay resolved by the Tukey criterion is calculated (Appendix B). Because the rotors were run in order of increasing texture, the run means are also refitted with a linear time term (Section 8).

Dependence on wind speed. A two-way ANOVA with runs nested in rotors tests the rotor $\times$ set point interaction against the residual (run $\times$ set point within rotor), that is, whether the ratio between rotors changes with wind speed by more than the scatter of repeated runs (Appendix B). The residual scatter differs between set points, so the F test is approximate; it is repeated without the two set points with the largest scatter.

Open-circuit voltage and source resistance. At each set point a straight line $V = V_{oc} - IR_{int}$ (a Thévenin equivalent) is fitted to the dwells above the voltage floor ($r^2 \geq 0.965$). The fit separates a change in generated voltage from a change in the voltage drop under load. V_{oc} , the zero-current intercept, is an extrapolated open-circuit voltage, not a measured one: from 1100 rpm its mean over all runs lies 0.5–2.7% below V_1 (single runs up to 4.1%), so the measured voltage falls faster over the first dwells than the fitted line. R_{int} is an apparent source resistance: it includes the winding, the rectifier with its commutation drop, the wiring, and the slowing of the rotor as the load rises. For Plain, R_{int} falls from $91\ \Omega$ at 10.2 m/s to $36\ \Omega$ at 38.0 m/s. A winding or wiring resistance does not fall with speed, and the commutation drop of a diode bridge rises with frequency [7, ch. 5]. We therefore attribute the fall to the rotor slowing under load and to the diodes, whose incremental resistance falls as the ladder current rises; the diode type was not recorded, so the two are not separated. As a check of the linear model, $V_{oc}^2/4R_{int}$, the maximum power of the fitted source, is 0.91–1.04 times P_{max} over all ladders. Both parameters are compared between rotors as P_{max} is.

Rotor speed. $\ln n_0$ at fan set points 600–1800 rpm (12.4–38.0 m/s) is compared between rotors as $\ln P_{max}$ is ($\sigma = 0.34\%$). Where both runs of a rotor give the same quantised value (29 of 52 cases), the quantisation error is shared and not in σ , so the n_0 intervals may be too narrow; V_1 , which is continuous, resolves the same six pairs. $\ln V_1$ and the voltage per rpm, $\ln(V_1/n_0)$ and $\ln(V_{oc}/n_0)$, are compared in the same way. The light-load tip-speed ratio is $\lambda_0 = 2\pi n_0 R/(60v)$ (n_0 in rpm, R in m, v in m/s). It exceeds the tip-speed ratio at peak power, where the load slows the rotor, and it uses the attachment radius R , so the ratio at the larger outer radius is higher.

Power coefficient. $C_{P,el} = P_{max}/(\frac{1}{2}\rho A v^3)$, with standard air ($\rho = 1.204\text{ kg/m}^3$), normalises power for wind speed. It includes generator and rectifier losses and is not the aerodynamic power coefficient.

6 Results

6.1 Power curves

Peak electrical power (geometric mean of each rotor’s two runs) rises from 0.030–0.034 W at 10.2 m/s to 3.67–4.53 W at 38.0 m/s (Fig. 4a), and $C_{P,el}$ rises with wind speed for every rotor, from 0.09% to 0.28% (Fig. 4b). With the measured-points-only calibration, $C_{P,el}$ at 38.0 m/s would be 17% higher.

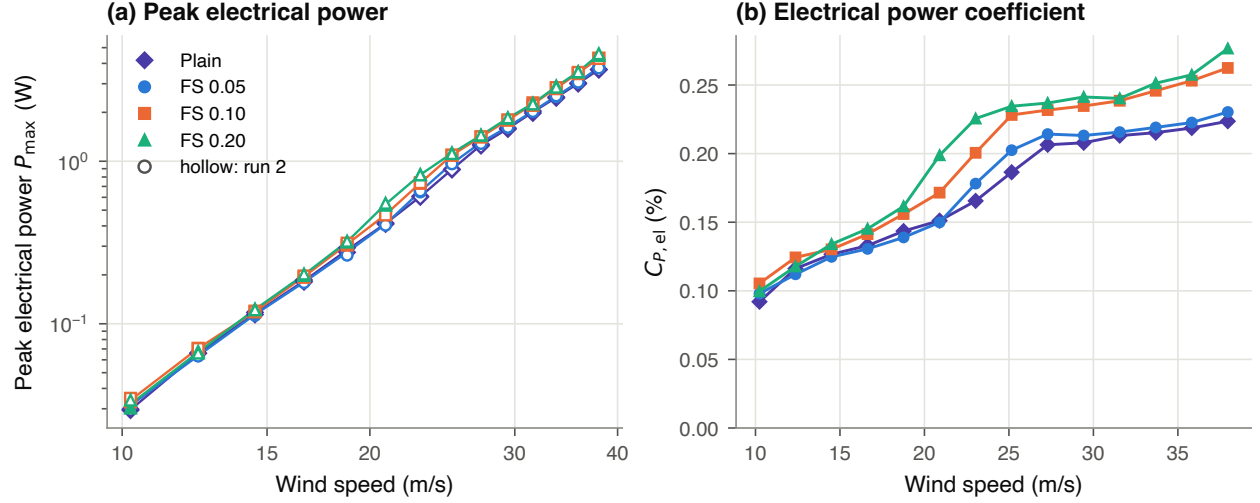


Figure 4. (a) Peak electrical power of every run on logarithmic axes (filled: run 1; hollow: run 2); lines join the geometric means of the two runs. (b) Electrical power coefficient, standard air density.

6.2 Rotor comparison

Averaged over wind speed, FS 0.10 and FS 0.20 produced +12.8% [+10.1, +15.5] and +16.2% [+13.5, +19.0] more peak power than Plain, and FS 0.05 +1.8% [−0.6, +4.2] (Table 3, Fig. 5). The rotor effect exceeds run-to-run variation ($F_{3,4} = 324$, $p < 0.001$; permutation $p = 1/105$). Each rotor’s two run means agree within 1%, and their scatter is not detectably larger than the point-to-point scatter within runs ($F_{4,52} = 1.39$, $p = 0.25$). Five of the six pairwise differences are resolved; FS 0.05 vs Plain is not. The smallest resolved difference, FS 0.20 over FS 0.10 (+3.0% [+0.6, +5.5]), keeps a lower limit of only +0.03% with the parabolic-fit estimator ($p = 0.048$). These intervals measure repeatability on one mounting. The differences of FS 0.10 and FS 0.20 from Plain would stay resolved for a mounting variation τ up to 2.0% and 2.6%, those from FS 0.05 up to 1.7% and 2.3%, and FS 0.20 vs FS 0.10 only up to 0.3%.

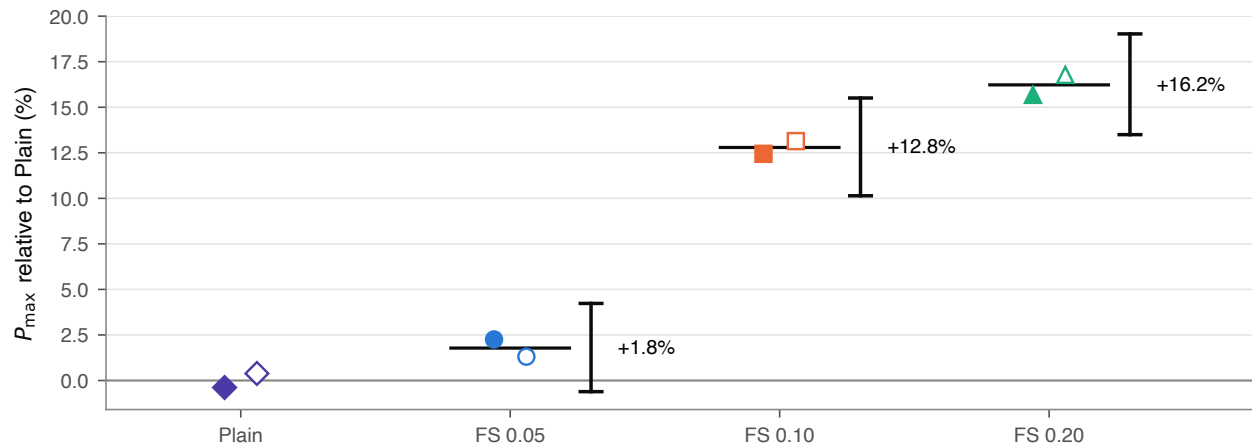


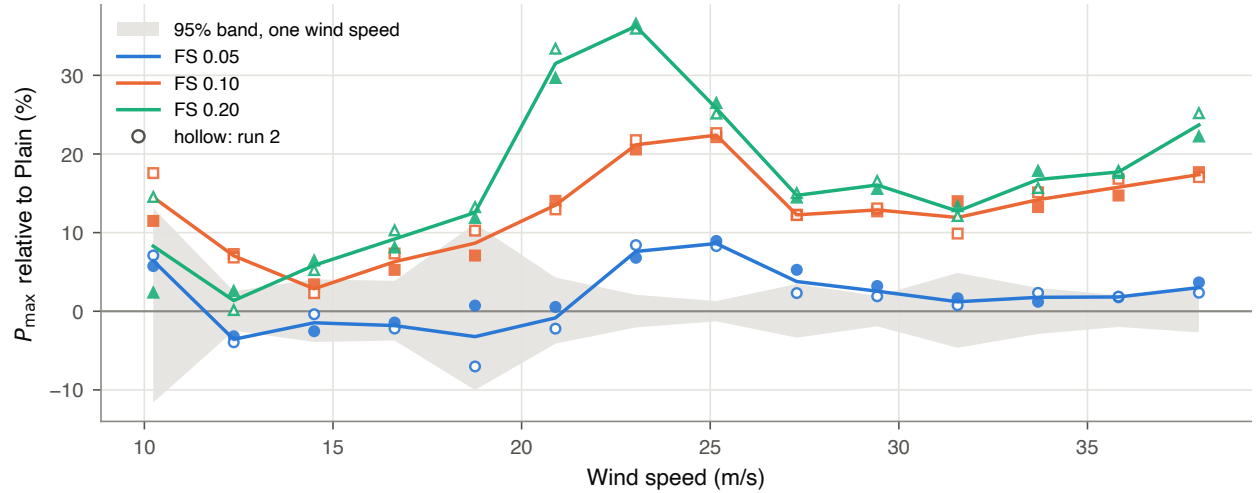
Figure 5. Each run’s mean $P_{\max}$ relative to the Plain rotor (filled: run 1; hollow: run 2). Bars: rotor mean with the Tukey simultaneous 95% interval for the difference from Plain.

Table 3. Pairwise comparisons of $P_{\max}$, geometric mean over wind speeds: Tukey simultaneous 95% intervals and adjusted p -values. Last column: the same for the parabolic-fit estimator.

Comparison	Change (%)	95% CI (%)	p	Parabolic fit (%)
FS 0.05 vs Plain	+1.8	[-0.6, +4.2]	$p = 0.12$	+1.6 [-1.2, +4.4]
FS 0.10 vs Plain	+12.8	[+10.1, +15.5]	$p < 0.001$	+13.0 [+10.0, +16.2]
FS 0.20 vs Plain	+16.2	[+13.5, +19.0]	$p < 0.001$	+16.2 [+13.1, +19.5]
FS 0.10 vs FS 0.05	+10.8	[+8.2, +13.5]	$p < 0.001$	+11.3 [+8.3, +14.4]
FS 0.20 vs FS 0.05	+14.2	[+11.5, +16.9]	$p < 0.001$	+14.4 [+11.3, +17.6]
FS 0.20 vs FS 0.10	+3.0	[+0.6, +5.5]	$p = 0.023$	+2.8 [+0.03, +5.7]

6.3 Dependence on wind speed

The rotor differences change with wind speed (Fig. 6). The rotor $\times$ set point interaction exceeds the scatter of repeated runs ($F_{39,52} = 9.2$, $p < 0.001$), also among the three textured rotors alone ($F_{26,39} = 7.6$, $p < 0.001$) and without the two set points with the largest scatter, 500 and 900 rpm ($F_{33,44} = 23.7$, $p < 0.001$). The gains are largest at 21–25 m/s (chord Reynolds number $6.6\text{--}8.0 \times 10^4$), the wind speeds at which FS 0.20 exceeds Plain by more than 25%: FS 0.20 peaks at +36.2% at 23 m/s, and FS 0.10 at +22.4% at 25 m/s. The steepest segment of $\ln P_{\max}$ against $\ln v$ (resolution 2.1 m/s, one set-point spacing) lies at 20 m/s for FS 0.20 and 22 m/s for FS 0.10 in both runs, at 22–24 m/s for FS 0.05 and at 24–26 m/s for Plain. The power of FS 0.10 and FS 0.20 therefore rises most steeply at a lower wind speed than in either Plain run; that of FS 0.05 overlaps Plain’s. The wind speeds carry the calibration uncertainty of Section 3.1; the ratios do not.

**Figure 6.** $P_{\max}$ of each textured rotor relative to the Plain rotor at each wind speed. Markers: individual runs (filled: run 1; hollow: run 2); lines: geometric mean of the two runs; band: 95% interval of a single-wind-speed difference from run-to-run scatter. All curves share the Plain reference, so features common to all three may come from it.

6.4 Open-circuit voltage and source resistance

Averaged over wind speed, V_{oc} of FS 0.05, FS 0.10 and FS 0.20 differs from Plain's by +1.2%, +6.4% and +8.5%, while R_{int} shows no resolved difference from Plain's (−0.1%, −0.1% and +0.2%; Table 4, Fig. 7). At fixed R_{int} the peak power of the fitted source scales with V_{oc}^2 , so the V_{oc} changes alone correspond to power changes of +2.4%, +13.1% and +17.7%, against +1.8%, +12.8% and +16.2% measured. R_{int} varies between rotors with wind speed ($F_{39,52} = 5.7$, $p < 0.001$): for FS 0.20 it is −7.1% at 21 m/s and +9.2% at 38 m/s relative to Plain, below Plain's from 15 to 25 m/s and above it from 27 m/s. FS 0.10 has the same sign at all 12 of these wind speeds, FS 0.05 at 9. Where the power gains are largest, the lower R_{int} supplies up to 43% of the gain in $\ln P_{max}$.

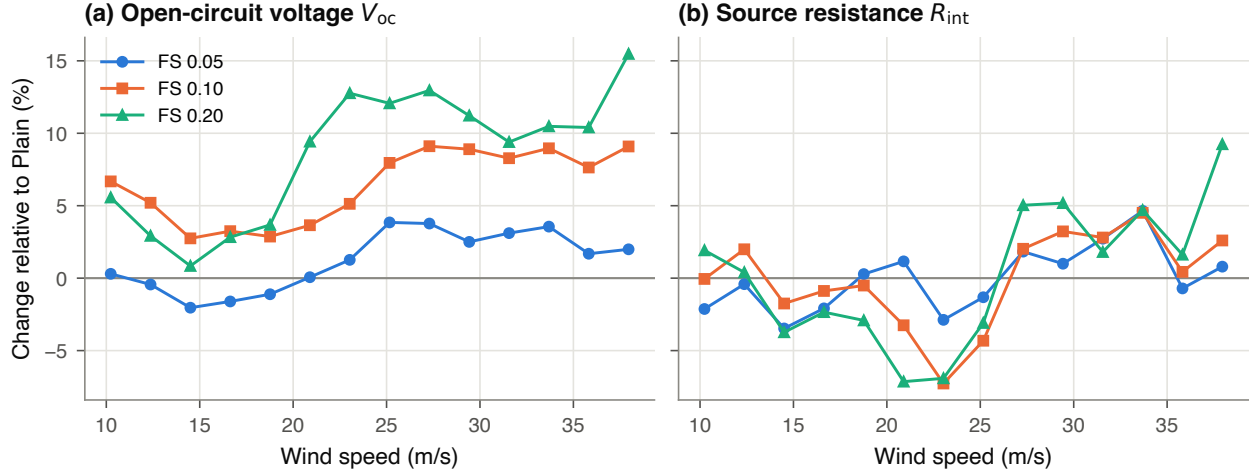


Figure 7. Thévenin parameters relative to the Plain rotor (geometric mean of two runs). (a) Extrapolated open-circuit voltage. (b) Apparent source resistance. All curves share the Plain reference, so features common to all three may come from it.

Table 4. Thévenin parameters and light-load speed n_0 relative to Plain, geometric mean over wind speeds (n_0 at fan set points 600–1800 rpm), with Tukey simultaneous 95% intervals; $C_{P,el}$ (standard air) and λ_0 (on R) at 38.0 m/s.

Rotor	Change relative to Plain (%)			At 38.0 m/s	
	V_{oc}	R_{int}	n_0	$C_{P,el}$ (%)	λ_0
Plain	reference	reference	reference	0.22	0.21
FS 0.05	+1.2 [+0.2, +2.2]	−0.1 [−1.5, +1.4]	+1.5 [+0.1, +2.9]	0.23	0.21
FS 0.10	+6.4 [+5.3, +7.4]	−0.1 [−1.5, +1.4]	+6.6 [+5.1, +8.1]	0.26	0.22
FS 0.20	+8.5 [+7.4, +9.6]	+0.2 [−1.3, +1.6]	+9.1 [+7.6, +10.6]	0.28	0.24

6.5 Rotor speed

From 12.4 to 38.0 m/s the light-load speed rises from 148–158 to 732–847 rpm (rotor means; Table 8) and λ_0 from 0.13–0.14 to 0.21–0.24, a rise of 61–81% (Fig. 8a). Averaged over wind speed, FS 0.10 and FS 0.20 turn +6.6% [+5.1, +8.1] and +9.1% [+7.6, +10.6] faster than Plain (rotor effect $F_{3,4} = 296$, $p < 0.001$; Table 4), and FS 0.20 turns +2.4% [+1.0, +3.8] faster than FS 0.10. FS 0.05 is only just resolved from Plain, +1.5% [+0.1, +2.9]; on the continuous V_1 its change is +1.9% [+1.2, +2.5]. Over the same set points V_1 changes by +1.9%, +6.5% and +9.8% and V_{oc} by +1.3%,

+6.3% and +8.7%; each speed change equals the corresponding V_1 and V_{oc} changes within 0.7 percentage points. Neither voltage per rpm shows a resolved difference between rotors: the largest difference is +0.7% [−0.7, +2.1] in V_1/n_0 (FS 0.20 vs FS 0.10) and −0.4% [−1.9, +1.2] in V_{oc}/n_0 (FS 0.20 vs Plain; Fig. 8c). The speed gain also changes with wind speed ($F_{36,48} = 6.7$, $p < 0.001$; Fig. 8b): up to 18.8 m/s, FS 0.10 and FS 0.20 are 2–7% and 1–6% faster than Plain; from 20.9 m/s, where the band of largest power gains begins, 4–11% and 9–16%.

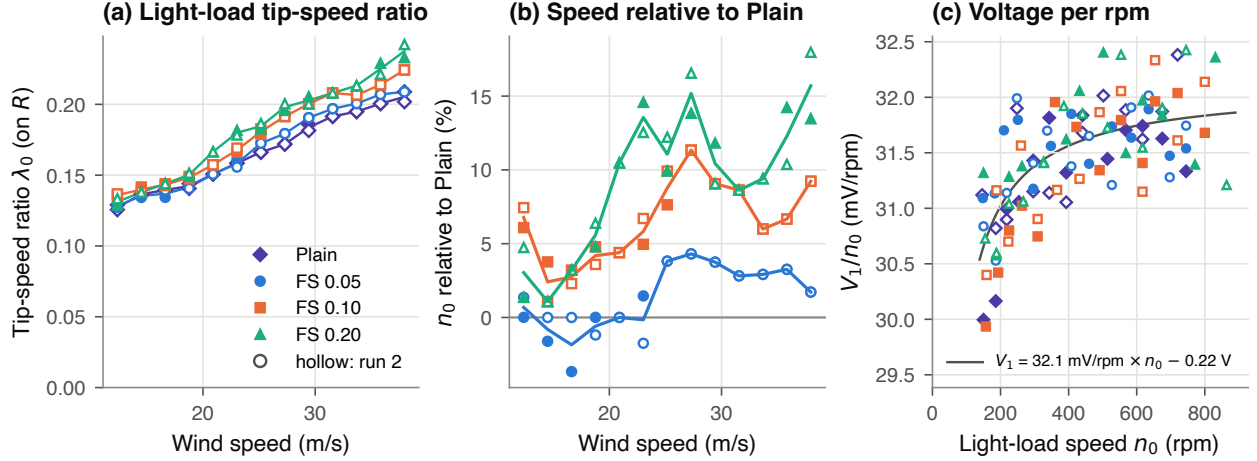


Figure 8. Light-load rotor speed n_0 from the tachometer at fan set points 600–1800 rpm (filled: run 1; hollow: run 2; lines: geometric mean of the two runs). (a) Tip-speed ratio λ_0 on the attachment radius. (b) n_0 of each textured rotor relative to Plain; one quantisation step is 0.7–4.0% of n_0 . (c) First-step voltage per rpm, V_1/n_0 , with the ratio implied by the least-squares line $V_1 = a + bn_0$.

6.6 Surface roughness and power

Neither Pa nor Ra orders the rotors by peak power (Table 5). Plain, with a regular 0.20 mm layer-line pattern, has the highest Ra (12.7 μm) and the lowest mean power. FS 0.05 and FS 0.10 share a 0.10 mm layer period and differ by 0.7 μm in Pa (13.9 and 13.2 μm) and 1.2 μm in Ra (9.4 and 8.2 μm), less than the two FS 0.20 fields differ from each other (4.8 μm in Pa, 2.0 μm in Ra); yet their power differs by +10.8% [+8.2, +13.5]. Pa roughly doubles from FS 0.10 to FS 0.20 (ratio 2.2), as the thickness setting does, but not from FS 0.05 to FS 0.10 (ratio 0.95). Mean peak power follows the nominal fuzzy-skin thickness, although FS 0.05 is not resolved from Plain. Ra is also not directly comparable between sets of different spatial structure: the 0.25 mm cut-off keeps about 65% of the amplitude of Plain’s layer lines, and the cut-off recommended for FS 0.20 exceeds the field (Appendix C).

7 Discussion

Where the power difference arises. We infer that the power difference arises in the rotor, not in the generator. Averaged over wind speed, the gain comes with a higher V_{oc} and no resolved change in R_{int} (Section 6.4). The light-load speed rises by the same fraction as V_{oc} , and neither voltage per rpm shows a resolved difference between rotors (Section 6.5). The generator therefore converts speed to voltage alike for every rotor, to within 2.1%, and the textured rotors turn faster. With the same V – I slope and a higher V_{oc} , they also turn faster at equal load current, that is, at about equal generator torque. The data do not separate a higher aerodynamic torque from a lower resisting

Table 5. Surface roughness and peak electrical power by rotor. Pa and Ra: profilometer (Section 2.3); FS 0.20 is the mean of two fields.

Rotor	Fuzzy skin (mm)	Pa (μm)	Ra (μm)	P_{max} change ^a (%)	P_{max} at 38.0 m/s (W)
Plain	none	19.0	12.7	reference	3.67
FS 0.05	0.050	13.9	9.4	+1.8 [−0.6, +4.2]	3.78
FS 0.10	0.101	13.2	8.2	+12.8 [+10.1, +15.5]	4.30
FS 0.20	0.202	28.7	10.3	+16.2 [+13.5, +19.0]	4.53

^aGeometric mean over wind speeds of the ratio to Plain, both runs, with the Tukey simultaneous 95% interval from run-to-run repeatability on one mounting. P_{max} at 38.0 m/s: geometric mean of the two runs.

torque. All rotors used the same generator and shaft bearings, so a lower resisting torque would have to come from the blade sets (mass, balance or mounting, none measured; the only documented difference is the FS 0.10 sensor magnet of Section 4.3) or from a change in the shared parts between rotors, which the run order does not exclude (Section 8). The R_{int} pattern that FS 0.10 and FS 0.20 share (Section 6.4) may come from the Plain reference rather than from the texture.

Wind-speed dependence. The gains of FS 0.10 and FS 0.20 peak at chord Reynolds numbers of $6.6\text{--}8.0 \times 10^4$, their power rises most steeply at a lower wind speed than Plain’s (Section 6.3), and their light-load speed gain is larger from 20.9 m/s (Section 6.5). A flow transition that depends on Reynolds number and that texture moves to a lower wind speed could produce this pattern. On a circular cylinder, roughness lowers the critical Reynolds number, leaves the subcritical drag unchanged and raises the transcritical drag [3], so its effect depends on the Reynolds-number range. The flow on these blades was not examined, so this remains a hypothesis.

Measured roughness. Ra and Pa do not rank the rotors by power (Section 6.6). Four explanations remain open: the realised texture of FS 0.05 and FS 0.10 may not follow the setting; one field per set may not resolve it; Pa and Ra may not capture the surface property that matters for this flow; or the power differences may not come from texture. With one print per level, the agreement between power and nominal thickness is an association (Section 8).

Operating regime and losses. The light-load tip-speed ratio does not exceed 0.24 (Section 6.5). A blade driven by drag alone stops producing driving force when it moves as fast as the wind, so this range is consistent with drag-driven operation. λ_0 also rises by 61–81% between 12.4 and 38.0 m/s. If the aerodynamic torque coefficient depended on λ alone, a rotor without resisting torque would run unloaded at a fixed λ . The rise is therefore consistent with a resisting torque (bearings and generator no-load losses) that is a larger share of the aerodynamic torque at low wind speed; a dependence of the aerodynamic torque on Reynolds number could also contribute. The largest $C_{P,\text{el}}$, 0.28%, is 18–108 times below the maximum averaged power coefficients of about 0.05–0.30 reported for Savonius rotors [8]. The comparison is indicative only: this rotor differs from a Savonius rotor in geometry, and $C_{P,\text{el}}$ includes generator and rectifier losses. If those losses were equal for all rotors at a given wind speed, a fractional change in aerodynamic power would appear magnified in P_{max} , and the measured gains would overstate the aerodynamic change. Speed-dependent losses would, however, be larger for the faster textured rotors, which acts the other way; the data do not separate the two.

8 Uncertainty and limitations

The Tukey intervals contain only run-to-run repeatability on one mounting. Two further sources are confounded with texture:

- **Print and mounting.** Each texture level is one print mounted once, so print-to-print and mounting variation are confounded with texture; Section 6.2 gives the mounting variation each difference could absorb. Plain also differs in layer period (0.20 mm against 0.10 mm) and has no slicer record. FS 0.05 and FS 0.10 share a 0.10 mm layer period, so their difference (+10.8% [+8.2, +13.5]) is the comparison least affected by layer height.
- **Run order.** The rotors were run in order of increasing texture, so slow drift is confounded with texture. A fit of the eight run means with a rotor term and a linear time term gives a drift of +0.8% per hour (95% interval -1.7% to $+3.3\%$). With that term FS 0.10 and FS 0.20 remain resolved from Plain (+11.5% [+7.0, +16.1] and +14.5% [+8.9, +20.4]; t intervals, 3 degrees of freedom), while FS 0.05 is +0.9% $[-2.3, +4.2]$. Under a linear-drift model the FS 0.10 and FS 0.20 differences therefore persist, while the FS 0.05 difference could be drift. The slope rests almost entirely (94% of the leverage) on the two Plain runs, the only pair an hour apart, and a change between rotors, rather than a steady drift, cannot be separated from a rotor effect.

Table 6 lists the other sources and their effect.

Table 6. Other sources of uncertainty.

Source	Size	Effect
Run-to-run repeatability	$\sigma = 0.58\%$ (run mean); 1.9% per point	In every interval.
Peak estimator	Changes within 0.5 percentage points	Same pairs resolved; FS 0.20 vs FS 0.10 keeps a lower limit of only +0.03% (Table 3).
Ramp dynamics	1 s dwells, equilibrium not verified; at 500 rpm the voltage rose between the first two dwells in seven of eight runs	Common protocol, but a bias that depends on rotor inertia or torque-speed slope would not cancel.
Rotor speed	Light load only, not at 500 rpm; resolution 0.7–4.0% of n_0	Partly in the n_0 intervals (only where repeat runs differ); λ_0 also carries the wind-speed calibration and uses R .
Wind-speed calibration	2.8–4.6% high at 10–14 m/s; 5% above a line through the measured points only at 38.0 m/s (an extrapolation, not a bound)	None on the comparison (identical fan speeds); affects absolute v and $C_{P,el}$.
Air density	Not recorded	Any change during the session is confounded with run order (above); $C_{P,el}$ assumes standard air.
Blockage	Not corrected	Assumed alike for rotors of one geometry; not quantified (test-section size not documented).
Electrical measurement	Not quantified	A gain error cancels in the ratios between rotors (same instrument and settings).
Surface measurement	One 2.0×1.4 mm field per set (FS 0.20: two), location not recorded; one cut-off for all sets	Pa and Ra may not represent the blades; Ra understates Plain's layer lines and the FS 0.20 surface (Appendix C).

9 Conclusions

1. FS 0.10 and FS 0.20 produced +12.8% [+10.1, +15.5] and +16.2% [+13.5, +19.0] more peak electrical power than Plain, averaged over 10.2–38.0 m/s. The FS 0.05 difference, +1.8% [−0.6, +4.2], was not resolved and could result from session drift.
2. The gains depended on wind speed and were largest at 21–25 m/s, where FS 0.20 reached +36.2%. The power of FS 0.10 and FS 0.20 rose most steeply at 22 m/s and 20 m/s, below Plain’s 24–26 m/s.
3. Averaged over wind speed, the gains came with a higher extrapolated open-circuit voltage and no resolved change in apparent source resistance; where they were largest, a lower source resistance also contributed. At light load FS 0.10 and FS 0.20 turned +6.6% [+5.1, +8.1] and +9.1% [+7.6, +10.6] faster than Plain, with no resolved difference in voltage per rpm. We infer that the difference arises in the rotor, not in the generator.
4. Ra and Pa from one field per set did not rank the rotors by power and do not confirm that the realised texture of FS 0.05 and FS 0.10 differs.
5. The intervals reflect repeatability on one print and one mounting per texture, and Plain also differs in layer period. The FS 0.10 and FS 0.20 differences from Plain would remain resolved for mounting variation up to 2.0–2.6% and under a linear-drift fit; attributing them to fuzzy skin requires replicate prints, remounting and an interleaved run order. FS 0.10 over FS 0.05, +10.8% [+8.2, +13.5], is the comparison least affected by layer height.

10 Recommendations

1. **Replicate and randomise.** Print at least two sets per thickness, Plain included, from one saved project (0.1 mm layers, 0.2 mm nozzle, one printer and filament lot; Print Lab or in house), in random order. Keep the project file and print record, and weigh and label every blade, including the present sets, so that mass and balance, the alternative to texture in Section 7, are known. Mount each set twice and run in an interleaved order, re-running Plain after every third rotor. This measures print and mounting variation and breaks the run-order confound.
2. **Log rotor speed at every dwell.** Connect a once-per-revolution sensor to the rig’s pulse input, or put T. Kang’s acquisition on the rig clock, and compute speed as revolutions over elapsed time rather than the mode of single-revolution speeds. T. Kang’s per-sample records of the present runs, not yet retrieved, could give speed at every dwell if they can be aligned to the rig’s time stamps. Per-dwell speed gives $C_{P,el}(\lambda)$ along each ladder and separates the electrical source impedance (winding, rectifier, commutation) from the rotor slowing. Also log the settled 0 A voltage, which the rig measures but does not save, measure the generator’s winding resistance and identify the rectifier diodes. Separating aerodynamic from resisting torque also requires a spin-down or motored friction test of each mounted rotor.
3. **Extend the wind calibration.** Measure air speed directly over the test range, record air temperature and pressure on every run, and document the test-section dimensions for a blockage estimate.
4. **Characterise the surfaces at fixed stations:** several fields per set on two blades and both faces, profiles along and across the layer lines, and evaluation lengths of at least five cut-off lengths.

10.1 Toward the fault-testing plan

Prof. Jeong asked for a fault-testing plan agreed before the next round. The rig provides an automated sweep of about 8 minutes with a run-to-run repeatability of 0.58% in the run mean and 1.9% at a single set point, measured on repeat runs of one mounting. Blade sets printed with different settings differed by up to +16.2% in mean peak power (Section 6.2), so fault blades and their reference set should be printed from one saved project, and the reference re-run between fault cases. Whether V_{oc} and R_{int} separate rotor-side from electrical faults is untested. R_{int} also reflects the rotor slowing under load (Section 5), so a rotor-side fault may change it as an electrical fault would; the first fault cases should test this with rotor speed logged at every dwell. Candidate faults are a tip mass on one blade (imbalance), a chipped, shortened or missing blade, added bearing drag, and an open or high-resistance generator phase. A short meeting to agree the fault list, severity levels and acceptance criteria should precede any fault prints.

Acknowledgement

T. Richards modelled the blade, A. Muszynski helped set up the fuzzy-skin print settings, and the College of Engineering's I²(s) Print Lab printed the blade sets.

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A Peak power and rotor speed by run and set point

Table 7. Peak electrical power $P_{\max}$ (W), largest dwell, every run. Runs 1 and 2 share the rotor’s mounting.

Fan rpm	v m/s	Plain		FS 0.05		FS 0.10		FS 0.20	
		1	2	1	2	1	2	1	2
500	10.2	0.0294	0.0298	0.0313	0.0317	0.0330	0.0348	0.0303	0.0339
600	12.4	0.0658	0.0660	0.0638	0.0633	0.0707	0.0704	0.0676	0.0660
700	14.5	0.1140	0.1175	0.1128	0.1153	0.1197	0.1184	0.1232	0.1218
800	16.6	0.1858	0.1811	0.1808	0.1794	0.1931	0.1969	0.1983	0.2023
900	18.8	0.2753	0.2937	0.2864	0.2644	0.3045	0.3135	0.3181	0.3220
1000	20.9	0.4162	0.4109	0.4158	0.4044	0.4716	0.4671	0.5362	0.5516
1100	23.0	0.6093	0.6031	0.6474	0.6572	0.7309	0.7381	0.8278	0.8240
1200	25.2	0.8894	0.8914	0.9700	0.9642	1.0875	1.0921	1.1261	1.1142
1300	27.3	1.2706	1.2463	1.3247	1.2875	1.4130	1.4123	1.4403	1.4474
1400	29.4	1.5788	1.5975	1.6391	1.6184	1.7898	1.7959	1.8351	1.8511
1500	31.6	1.9799	2.0368	2.0413	2.0236	2.2892	2.2066	2.2762	2.2513
1600	33.7	2.4530	2.4832	2.4978	2.5259	2.7952	2.8414	2.9085	2.8549
1700	35.8	3.0003	3.0273	3.0671	3.0700	3.4574	3.5213	3.5436	3.5516
1800	38.0	3.6640	3.6666	3.7998	3.7517	4.3137	4.2905	4.4798	4.5882

Table 8. Light-load rotor speed n_0 (rpm), every run: T. Kang’s most frequent per-revolution speed at each set point (Section 3.3). At 500 rpm (in parentheses) the fan settled with the load at 10 mA; those values are not used.

Fan rpm	v m/s	Plain		FS 0.05		FS 0.10		FS 0.20	
		1	2	1	2	1	2	1	2
500	10.2	(92)	(90)	(91)	(91)	(96)	(96)	(96)	(95)
600	12.4	150	146	148	150	157	159	150	155
700	14.5	186	186	183	186	193	188	188	188
800	16.6	218	218	210	218	225	223	225	225
900	18.8	254	248	251	248	263	260	263	267
1000	20.9	296	296	296	296	309	309	327	327
1100	23.0	343	343	348	337	360	366	393	386
1200	25.2	393	393	408	408	423	432	432	441
1300	27.3	441	441	460	460	491	491	502	514
1400	29.4	514	502	527	527	554	554	568	554
1500	31.6	568	568	584	584	617	617	617	617
1600	33.7	617	617	635	635	654	654	675	675
1700	35.8	675	675	697	697	720	720	771	745
1800	38.0	745	720	745	745	800	800	831	864

B Statistical details

Model. With $y_{rjs} = \ln P_{\max}$ for rotor r (4), run j (2) and set point s (14),

$$y_{rjs} = \mu + \alpha_r + \beta_s + (\alpha\beta)_{rs} + u_{rj} + e_{rjs},$$

where u_{rj} is a run offset within the rotor and e_{rjs} the residual [4]. The rotor effect is tested against run within rotor, which is the one-way ANOVA on run means. The interaction $(\alpha\beta)_{rs}$ and the run term are tested against the residual (Table 9). The run offsets u_{rj} cancel from both the interaction and the residual mean squares, so the interaction test needs no assumption about them; it does assume equal residual variance, so it is approximate here.

Table 9. Analysis of variance of $\ln P_{\max}$. MS in units of 10^{-4} .

Source	df	MS	F	p	Tested against
Rotor	3	1 550	324	$p < 0.001$	run within rotor
Set point (wind speed)	13	187 000			
Rotor $\times$ set point	39	31.7	9.22	$p < 0.001$	residual
Run within rotor	4	4.78	1.39	$p = 0.25$	residual
Residual	52	3.44			

Comparisons. With run means $\bar{y}_{rj}$, rotor means $\bar{y}_r$ and pooled $\sigma^2 = \sum_r \sum_j (\bar{y}_{rj} - \bar{y}_r)^2 / 4$, Tukey’s interval for rotors a and b is $\exp(\bar{y}_b - \bar{y}_a \pm q_{0.05;4,4} \sigma / \sqrt{2}) - 1$, a half-width of 0.024 in $\ln P_{\max}$ (about 2.4%). The same procedure is applied to $\ln V_{oc}$ and $\ln R_{int}$, and to $\ln n_0$, $\ln V_1$, $\ln(V_1/n_0)$ and $\ln(V_{oc}/n_0)$ at fan set points 600–1800 rpm. The permutation test enumerates all 105 ways of splitting the eight run means into four pairs; for $\ln n_0$ it also gives $p = 1/105$.

Mounting variation. If each rotor’s mean carried an independent mounting offset with standard deviation τ , the difference of two rotor means would have variance $\sigma^2 + 2\tau^2$. It stays resolved by the Tukey criterion while $|\bar{y}_b - \bar{y}_a| > (q_{0.05;4,4} / \sqrt{2}) \sqrt{\sigma^2 + 2\tau^2}$, which gives the values of τ in Section 5.

Drift. The run means are fitted as $\bar{y}_{rj} = a_r + \beta t_{rj}$, with t_{rj} the start time of the run in hours, by least squares (3 residual degrees of freedom).

Residual scatter. The per-point residual standard deviation is 1.9% overall; it is largest at 500 rpm (4.1%) and smallest at 1400 rpm (0.2%).

C Surface analysis

The Keyence export is a height map in millimetres, quantised to 1 μm ; each image column is one spanwise profile. A column is dropped if more than 10% of its points are missing, or more than 2% between its first and last valid points: 1.8% of columns or fewer for Plain, FS 0.05 and FS 0.10, and 5.1% for FS 0.20, where missing points concentrate on steep slopes, so its values may be understated. Remaining interior gaps are filled linearly, a least-squares line is removed, and the profiles of a field are cut to the length of the shortest. P_a is the mean absolute deviation of the levelled profile. For R_a , the levelled profile minus its Gaussian mean line (ISO 16610-21 [5]; $\alpha = \sqrt{\ln 2/\pi}$, $\lambda_c = 0.25 \text{ mm}$) is evaluated with $\lambda_c/2$ discarded at each end, where the filter has end effects. The layer period is the peak of the mean power spectrum between 60 and 320 μm . For periodic profiles ISO 4288 [6] selects $\lambda_c = 0.25 \text{ mm}$ for spacings of 0.04–0.13 mm, which fits FS 0.05 and FS 0.10; the Plain set’s 0.2 mm period would call for 0.8 mm, and the non-periodic FS 0.20 surface (R_a above 10 μm) for 8 mm, both longer than the 1.4 mm field. The filter transmits 50% at λ_c , so R_a keeps about 65% of the amplitude of the Plain set’s 0.20 mm layer lines and 99% of the 0.10 mm ones; Plain’s R_a is therefore understated relative to FS 0.05 and FS 0.10. P_a is unfiltered.

D Wind-speed calibration

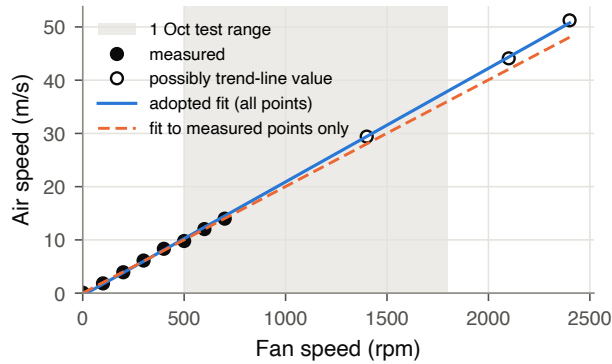


Figure 9. Tunnel calibration, Test 1 (13 February 2026). Filled: measured. Hollow: marked in the calibration file as possibly read from the trend line of the original record (“trend line?” in Table 10). Shaded: the range of this test.

Table 10. Calibration points and the adopted fit $v = 0.02132N - 0.424$. A line through the measured points alone has a residual standard deviation of 0.19 m/s.

Fan (rpm)	v (m/s)	Source	Fit (m/s)	Fit – meas. (m/s)
0	0.08	measured	−0.42	−0.50
100	1.83	measured	1.71	−0.12
200	3.92	measured	3.84	−0.08
300	6.12	measured	5.97	−0.15
400	8.33	measured	8.10	−0.23
500	9.79	measured	10.24	+0.45
600	12.03	measured	12.37	+0.34
700	13.99	measured	14.50	+0.51
1400	29.40	trend line?	29.43	+0.03
2100	44.10	trend line?	44.35	+0.25
2400	51.24	trend line?	50.75	−0.49

E Data package and reproducibility

The accompanying archive holds the raw files of the eight runs (`1_rig_sweeps/`), T. Kang’s rotor-speed summaries and his script `RPM.m` (`2_rotor_speed/`), the profilometer height maps (`3_surface_scans/`), the analysis-ready tables behind this report (`4_derived/`), the rotor geometry, blade mesh, wind calibration and slicer summary (`5_reference/`), the data figures, Figs 1 and 3–9 (`6_figures/`), and the analysis code (`7_code/`). `README.md` explains the layout, `DATA_DICTIONARY.md` defines every column, and `MANIFEST.csv` gives each file’s SHA-256 checksum. `7_code/build_report.py` (Python 3) regenerates every result, table and data figure in this report from the raw files.